

Midterm Exam Solutions

GSE 510

Total Points: 100
10 Questions, 10 Points Each

Instructions: These are worked solutions for the midterm exam.

1. Sets, Bounds, and Economic Feasibility

(10 points)

The feasible set is

$$X = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, 2x_1 + x_2 \leq 10\}.$$

(a) For $(2, 3)$,

$$2(2) + 3 = 4 + 3 = 7 \leq 10.$$

Also, $2 \geq 0$ and $3 \geq 0$. Therefore,

$$(2, 3) \in X.$$

(b) For $(5, 1)$,

$$2(5) + 1 = 10 + 1 = 11 > 10.$$

Therefore,

$$(5, 1) \notin X.$$

(c) For $(0, 10)$,

$$2(0) + 10 = 10 \leq 10.$$

Also, both components are nonnegative. Therefore,

$$(0, 10) \in X.$$

(d) To find the largest possible value of x_1 , set $x_2 = 0$. Then

$$2x_1 \leq 10,$$

so

$$x_1 \leq 5.$$

Therefore, the largest possible value of x_1 is

$$5.$$

(e) To find the largest possible value of x_2 , set $x_1 = 0$. Then

$$x_2 \leq 10.$$

Therefore, the largest possible value of x_2 is

$$10.$$

(f) Yes, X is bounded. Every feasible bundle satisfies

$$0 \leq x_1 \leq 5 \quad \text{and} \quad 0 \leq x_2 \leq 10.$$

Hence X lies inside the rectangle $[0, 5] \times [0, 10]$.

2. Logic and Proof

(10 points)

Consider the statement:

If $x > 2$, then $x^2 > 4$.

- (a) The proposition P is

$$P : x > 2.$$

The proposition Q is

$$Q : x^2 > 4.$$

- (b) The converse is:

If $x^2 > 4$, then $x > 2$.

- (c) The contrapositive is:

If $x^2 \leq 4$, then $x \leq 2$.

- (d) The original statement is true. If $x > 2$, then x is positive and larger than 2, so squaring gives

$$x^2 > 2^2 = 4.$$

- (e) The converse is false. For example, take $x = -3$. Then

$$x^2 = 9 > 4,$$

but

$$x = -3 \not> 2.$$

So $x^2 > 4$ does not imply $x > 2$.

- (f) The contrapositive is true because it is logically equivalent to the original statement. We can also check it directly: if

$$x^2 \leq 4,$$

then

$$-2 \leq x \leq 2,$$

so in particular

$$x \leq 2.$$

3. Sequences, Limits, and Closed Sets

(10 points)

Let

$$S = (0, 1]$$

and

$$x_n = \frac{1}{n}.$$

- (a) For every positive integer n ,

$$n \geq 1.$$

Hence

$$0 < \frac{1}{n} \leq 1.$$

Therefore,

$$x_n = \frac{1}{n} \in (0, 1] = S.$$

- (b) We have

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Therefore,

$$\lim_{n \rightarrow \infty} x_n = 0.$$

- (c) The limit does not belong to S , because

$$S = (0, 1]$$

excludes 0. Therefore,

$$0 \notin S.$$

- (d) The set S is not closed. The sequence $x_n = 1/n$ lies entirely in S , but it converges to 0, which is not in S . A closed set must contain the limits of all convergent sequences contained in the set.
- (e) The set S is not open in $\mathbb{R}$. For example, $1 \in S$, but any open interval around 1 contains points greater than 1, which are not in S . Thus, no small open ball around 1 is contained entirely in S .

4. Linear Algebra and a Market System

(10 points)

Demand and supply are

$$Q_d = 60 - 2P, \quad Q_s = 10 + 3P.$$

In equilibrium,

$$Q_d = Q_s = Q.$$

- (a) Since $Q = Q_d$ and $Q = Q_s$, we have

$$Q = 60 - 2P$$

and

$$Q = 10 + 3P.$$

Rearranging,

$$Q + 2P = 60$$

and

$$Q - 3P = 10.$$

- (b) With

$$\mathbf{x} = \begin{bmatrix} Q \\ P \end{bmatrix},$$

the system is

$$\begin{bmatrix} 1 & 2 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} Q \\ P \end{bmatrix} = \begin{bmatrix} 60 \\ 10 \end{bmatrix}.$$

- (c) The determinant is

$$|\mathbf{A}| = \begin{vmatrix} 1 & 2 \\ 1 & -3 \end{vmatrix} = 1(-3) - 2(1) = -3 - 2 = -5.$$

- (d) Since

$$|\mathbf{A}| = -5 \neq 0,$$

the matrix $\mathbf{A}$ is nonsingular.

- (e) From the two equations,

$$Q + 2P = 60$$

and

$$Q - 3P = 10.$$

Subtract the second equation from the first:

$$(Q + 2P) - (Q - 3P) = 60 - 10.$$

Thus,

$$5P = 50,$$

so

$$P^* = 10.$$

Then

$$Q^* = 60 - 2(10) = 40.$$

Therefore, the equilibrium price and quantity are

$$P^* = 10, \quad Q^* = 40.$$

5. Span, Linear Independence, and Interpretation

(10 points)

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

- (a) A general linear combination is

$$a\mathbf{v}_1 + b\mathbf{v}_2 = a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

Therefore,

$$a\mathbf{v}_1 + b\mathbf{v}_2 = \begin{bmatrix} a + 3b \\ 2a + 6b \end{bmatrix}.$$

- (b) The vectors are not linearly independent because

$$\mathbf{v}_2 = 3\mathbf{v}_1.$$

Therefore, they are linearly dependent.

- (c) Since $\mathbf{v}_1$ and $\mathbf{v}_2$ lie on the same line, they do not span $\mathbb{R}^2$. Their span is only the line through the origin generated by $\mathbf{v}_1$.

- (d) Yes. We want

$$\begin{bmatrix} a + 3b \\ 2a + 6b \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}.$$

Since

$$\begin{bmatrix} 4 \\ 8 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

one pair that works is

$$a = 4, \quad b = 0.$$

There are many others. For example, $a = 1$ and $b = 1$ also works because

$$\mathbf{v}_1 + \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}.$$

- (e) No. Any vector generated by $\mathbf{v}_1$ and $\mathbf{v}_2$ has the form

$$\begin{bmatrix} a + 3b \\ 2a + 6b \end{bmatrix} = (a + 3b) \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Thus, every generated vector has second component equal to twice the first component. The vector

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

does not satisfy this property, so it cannot be generated.

- (f) Economically, if two activity vectors do not span $\mathbb{R}^2$, then combinations of those activities cannot produce every possible output or input bundle in two dimensions. The technology is restricted to a lower-dimensional set of feasible directions.

6. OLS in Matrix Form

(10 points)

The model is

$$y_i = \beta_0 + \beta_1 x_i + u_i.$$

The data are

$$\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

(a) The model in matrix form is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}.$$

(b) We have

$$\mathbf{X}' = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Therefore,

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}.$$

(c) Also,

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}.$$

(d) The determinant of $\mathbf{X}'\mathbf{X}$ is

$$3(5) - 3(3) = 15 - 9 = 6.$$

Hence,

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}.$$

(e) The OLS estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Therefore,

$$\hat{\boldsymbol{\beta}} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 7 \\ 10 \end{bmatrix}.$$

Multiplying,

$$\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 7 \\ 10 \end{bmatrix} = \begin{bmatrix} 35 - 30 \\ -21 + 30 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}.$$

Thus,

$$\hat{\boldsymbol{\beta}} = \frac{1}{6} \begin{bmatrix} 5 \\ 9 \end{bmatrix} = \begin{bmatrix} 5/6 \\ 3/2 \end{bmatrix}.$$

So

$$\hat{\beta}_0 = \frac{5}{6}, \quad \hat{\beta}_1 = \frac{3}{2}.$$

(f) The fitted regression line is

$$\hat{y} = \frac{5}{6} + \frac{3}{2}x.$$

7. Differentiation and Optimization

(10 points)

Revenue and cost are

$$R(q) = 80q - 2q^2, \quad C(q) = 20 + 10q + q^2.$$

Profit is

$$\pi(q) = R(q) - C(q).$$

(a) We have

$$\pi(q) = 80q - 2q^2 - (20 + 10q + q^2).$$

Simplifying,

$$\pi(q) = 70q - 3q^2 - 20.$$

(b) The derivative is

$$\pi'(q) = 70 - 6q.$$

(c) The first-order condition is

$$70 - 6q = 0.$$

Solving,

$$6q = 70,$$

so

$$q^* = \frac{70}{6} = \frac{35}{3}.$$

(d) The second derivative is

$$\pi''(q) = -6.$$

(e) Since

$$\pi''(q) = -6 < 0,$$

the critical point is a local maximum.

(f) Maximum profit is

$$\pi\left(\frac{35}{3}\right) = 70\left(\frac{35}{3}\right) - 3\left(\frac{35}{3}\right)^2 - 20.$$

Compute:

$$70\left(\frac{35}{3}\right) = \frac{2450}{3}$$

and

$$3\left(\frac{35}{3}\right)^2 = 3\left(\frac{1225}{9}\right) = \frac{1225}{3}.$$

Hence,

$$\pi\left(\frac{35}{3}\right) = \frac{2450}{3} - \frac{1225}{3} - 20 = \frac{1225}{3} - 20.$$

Since

$$20 = \frac{60}{3},$$

we get

$$\pi\left(\frac{35}{3}\right) = \frac{1165}{3}.$$

8. Linear and Quadratic Approximation

(10 points)

The production function is

$$F(K) = 20K^{1/2}.$$

(a) The first derivative is

$$F'(K) = 20 \cdot \frac{1}{2} K^{-1/2} = 10K^{-1/2}.$$

The second derivative is

$$F''(K) = 10\left(-\frac{1}{2}\right) K^{-3/2} = -5K^{-3/2}.$$

(b) At $K = 100$,

$$F(100) = 20(100)^{1/2} = 20(10) = 200.$$

(c) The linear approximation around $K_0 = 100$ is

$$F(K) \approx F(100) + F'(100)(K - 100).$$

Now

$$F'(100) = 10(100)^{-1/2} = \frac{10}{10} = 1.$$

Since $109 - 100 = 9$,

$$F(109) \approx 200 + 1(9) = 209.$$

(d) The quadratic approximation is

$$F(K) \approx F(100) + F'(100)(K - 100) + \frac{1}{2}F''(100)(K - 100)^2.$$

First,

$$F''(100) = -5(100)^{-3/2}.$$

Since

$$100^{3/2} = 1000,$$

we get

$$F''(100) = -\frac{5}{1000} = -\frac{1}{200}.$$

Therefore,

$$F(121) \approx 200 + 1(21) + \frac{1}{2} \left(-\frac{1}{200} \right) (21)^2.$$

$$F(121) \approx 221 - \frac{441}{400}.$$

Thus,

$$F(121) \approx 219.9.$$

(e) The exact value is

$$F(121) = 220.$$

9. Implicit Differentiation and Indifference Curves

(10 points)

The indifference curve is

$$F(x, y) = x^{1/2} + y^{1/2} - 5 = 0.$$

(a) The partial derivatives are

$$F_x = \frac{1}{2}x^{-1/2}$$

and

$$F_y = \frac{1}{2}y^{-1/2}.$$

(b) By the implicit function theorem,

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

Therefore,

$$\frac{dy}{dx} = -\frac{\frac{1}{2}x^{-1/2}}{\frac{1}{2}y^{-1/2}}.$$

Canceling the $\frac{1}{2}$ terms,

$$\frac{dy}{dx} = -\frac{x^{-1/2}}{y^{-1/2}} = -\frac{\sqrt{y}}{\sqrt{x}}.$$

(c) At $(x, y) = (4, 9)$,

$$\frac{dy}{dx} = -\frac{\sqrt{9}}{\sqrt{4}} = -\frac{3}{2}.$$

(d) If x increases from 4 to 4.2, then

$$\Delta x = 0.2.$$

Using linear approximation,

$$\Delta y \approx \frac{dy}{dx} \Delta x.$$

At $(4, 9)$,

$$\frac{dy}{dx} = -\frac{3}{2}.$$

Hence,

$$\Delta y \approx -\frac{3}{2}(0.2) = -0.3.$$

Therefore, y must decrease by approximately 0.3. Starting from $y = 9$, the new approximate value is

$$y \approx 8.7.$$

- (e) The sign is negative because along an indifference curve, if the consumer receives more of good x , then they must give up some of good y to remain at the same utility level. Hence the indifference curve is downward sloping.

10. Comparative Statics and the Implicit Function Theorem

(10 points)

The firm maximizes

$$\pi(q, t) = 100q - \frac{1}{2}q^2 - tq.$$

- (a) The first-order condition is

$$\frac{\partial \pi(q, t)}{\partial q} = 100 - q - t = 0.$$

- (b) Solving the first-order condition,

$$100 - q - t = 0.$$

Therefore,

$$q^*(t) = 100 - t.$$

- (c) Differentiating the explicit solution,

$$\frac{dq^*}{dt} = -1.$$

- (d) Define

$$F(q, t) = 100 - q - t.$$

Then

$$F_q = -1$$

and

$$F_t = -1.$$

By the implicit function theorem,

$$\frac{dq^*}{dt} = -\frac{F_t}{F_q}.$$

Substituting,

$$\frac{dq^*}{dt} = -\frac{-1}{-1} = -1.$$

This agrees with the answer from the explicit solution.

- (e) If $t = 20$, then

$$q^*(20) = 100 - 20 = 80.$$

- (f) If the tax increases from 20 to 23, then

$$\Delta t = 3.$$

Using linear approximation,

$$\Delta q^* \approx \frac{dq^*}{dt} \Delta t.$$

Since

$$\frac{dq^*}{dt} = -1,$$

we get

$$\Delta q^* \approx -1(3) = -3.$$

Therefore,

$$q^*(23) \approx q^*(20) - 3 = 80 - 3 = 77.$$

In this case the approximation is exact, because $q^*(t)$ is linear.

(g) The derivative

$$\frac{dq^*}{dt} = -1$$

is negative. Economically, this means that an increase in the per-unit tax reduces the firm's optimal output. Here, a one-unit increase in the tax reduces optimal output by one unit.